Q26 Discussion Questions

1. InstructGPT's 1.3B model was preferred by humans over the 175B base model it was tuned from. If fine-tuning reshapes the response distribution rather than adding capability, where does the "improvement" live — in the model, or in the match between the model's distribution and the raters' expectations? What would an experiment that separates these look like?
2. LoRA constrains the update to rank r and EWC penalizes movement of important weights, yet both are described as "catastrophic forgetting mitigations." Are they protecting the same thing? Describe a task pair where you would expect the rank constraint to fail but an importance penalty to succeed.
3. Ovadia et al. found RAG beats unsupervised fine-tuning for injecting both seen and unseen knowledge. Yet domain-adapted models (legal, medical, code) clearly exist and help. What are those pipelines actually buying — knowledge, behavior, or vocabulary/tokenization fit — and how would you tell empirically?
4. The module argues naive online SGD destabilizes RLHF-trained models because it removes the KL anchor and compounds drift. Security and governance concerns aside, could an anchored online update (small-r LoRA refreshed per stream, reference model pinned) be made safe enough for production? What failure mode would you monitor first?
5. Given the rule "retrieve for facts, tune for behavior," when is tuning for facts nonetheless the right call? Consider latency, context-window cost, retrieval failure modes at scale, and the fact that some knowledge is only expressible as a skill rather than a lookup.